

Project Report in the Module
MHYWI05 “Statistical Learning for Earth System Sciences”

**Statistical Learning for Earth System Sciences:
Crop Yield Prediction and European Climate Trends**

Module Code:	MHYWI05
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1 Introduction

This report applies statistical learning methods to two environmental datasets with different structures and inferential goals. The first is a panel of Iowa crop yields (25 locations, 60 years) with monthly growing-season meteorological predictors; here the question is predictive, and specifically how the standard random train/test split compares to one that respects the data’s temporal structure. The second is gridded European climate data over 1950–2025; here the question is inferential, namely how strong the long-term temperature and precipitation trends are and how correcting for thousands of simultaneous tests affects which trends survive. Both analyses turn on assumptions that are easy to ignore: independence of observations in the crop panel, and independence of tests in the climate grids.

2 Iowa Crop Yield Data

2.1 Data and dimensionality: Answer 1a

The dataset contains $n = 1500$ observations (25 locations $\times$ 60 years) and $p = 23$ predictors: six monthly values each of precipitation, radiation, and temperature during the growing season, plus five extreme-event indicators (frost days, warm days, maximum and minimum temperature, and maximum 5-day precipitation). There are no missing values.

PCA on the standardised predictors shows that **11 components are required to explain at least 80% of the total variance** (81.3% cumulative; Figure 1). The scree plot shows no sharp elbow: variance declines gradually rather than concentrating in a few components, indicating moderate but not extreme redundancy among the predictors.

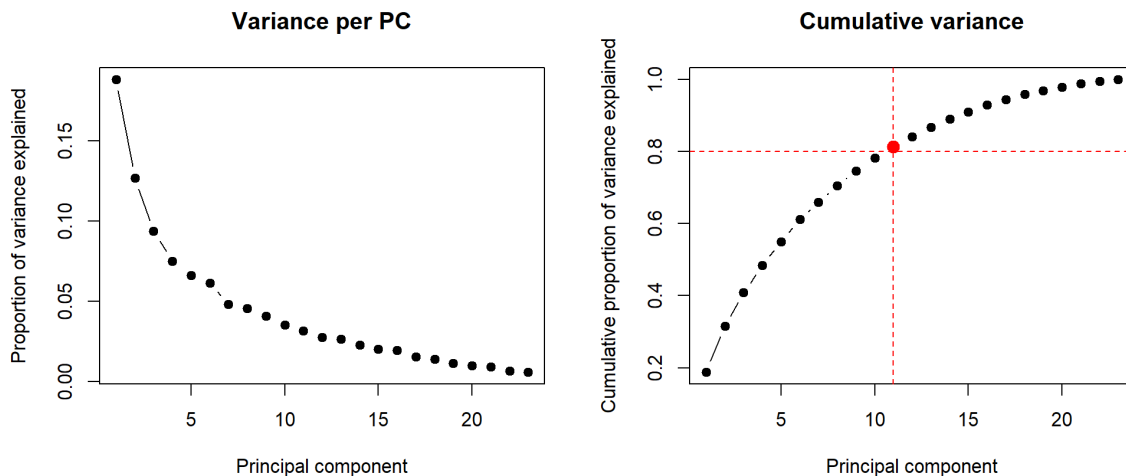


Figure 1: Variance explained per principal component (left) and cumulative variance (right), with the 80% threshold and the 11th component marked in red.

2.2 Violation of the i.i.d. assumption: Answer 1b

The data violate the independent and identically distributed assumption through strong cross-sectional (spatial) dependence. A representative monthly temperature predictor, `temp_3`, is correlated at $r = 0.958$ across the 25 Iowa locations within the same year, and yield itself shows a mean cross-location correlation of $r = 0.685$. Growing conditions are shared across the state, so observations from the same year are not independent draws.

A fixed-effects regression confirms this: fitting yield on location and year as categorical factors alone, with no weather information, explains **71.8% of the variance**. Temporal dependence is

comparatively weak — the mean within-location correlation between year and yield is $r = -0.094$ — so the identical-distribution assumption is a smaller concern than independence.

2.3 Random train/test split: Answer 1c

The data were split 50/50 ($n = 750$ each) using `set.seed(99)`. Two models were fit: best subset regression with size chosen by minimum BIC, and lasso with 5-fold cross-validation. For the lasso, both $\lambda_{\min}$ and λ_{1se} are reported. Results are shown in Table 1.

Model	Vars	Train MSE	Train R^2	Test MSE	Test R^2
Best subset (BIC, $k = 11$)	11	2.8381	0.6704	3.2581	0.6343
Lasso $\lambda_{\min}$	20	2.7716	0.6782	3.2366	0.6367
Lasso λ_{1se}	12	3.0566	0.6451	3.3540	0.6235

Table 1: Random 50/50 split ($n_{\text{train}} = n_{\text{test}} = 750$). Test R^2 is computed against the test set’s own mean.

λ_{1se} retains 12 variables against $\lambda_{\min}$ ’s 20, at a cost of only ~ 0.013 in test R^2 ; it is the more defensible choice for a report where interpretability matters.

Best subset (11 variables) selected: `pr_2`, `pr_3`, `rad_2`, `rad_3`, `temp_2`, `temp_3`, `temp_4`, `temp_6`, `warm_days`, `max_temp`, `max_5day_pr`.

Lasso at λ_{1se} (12 variables) selected a very similar set: `pr_2`, `pr_3`, `rad_2`, `rad_3`, `rad_5`, `temp_2`, `temp_3`, `temp_4`, `temp_6`, `warm_days`, `max_temp`, `max_5day_pr`.

The two sets differ only by `rad_5`, which lasso includes. Test R^2 for both methods lands in the 0.62–0.64 range, and the predicted-versus-observed plots in Figure 2 track the 1:1 line closely across the full yield range.

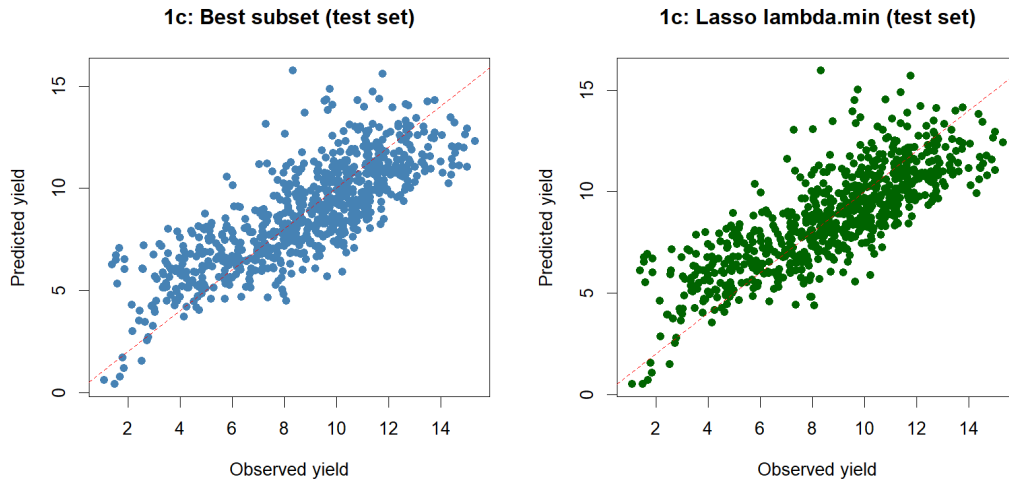


Figure 2: Predicted vs. observed yield on the test set (random split), best subset (left) and lasso $\lambda_{\min}$ (right); dashed line is 1:1.

2.4 Year-parity split and the effect of spatial leakage: Answer 1d

The models were refitted with an alternative split: even years for training, odd years for testing. This yields 750 observations in each set, but with no year appearing in both training and test data. Results are shown in Table 2.

Model	Vars	Train MSE	Train R^2	Test MSE	Test R^2
Best subset (BIC, $k = 15$)	15	2.7494	0.6934	4.5750	0.4648
Lasso $\lambda_{\min}$	23	2.6839	0.7007	4.7114	0.4489
Lasso λ_{1se}	17	2.9036	0.6762	4.3304	0.4934

Table 2: Year-parity split ($n_{\text{train}} = n_{\text{test}} = 750$).

Training performance is essentially unchanged; test performance drops substantially. Best subset test R^2 falls from 0.634 to 0.465, and lasso at $\lambda_{\min}$ falls from 0.637 to 0.449. Figure 3 shows the cause visually: the scatter around the 1:1 line is markedly wider, with predictions for high-yield observations routinely falling short by 2–4 t/ha.

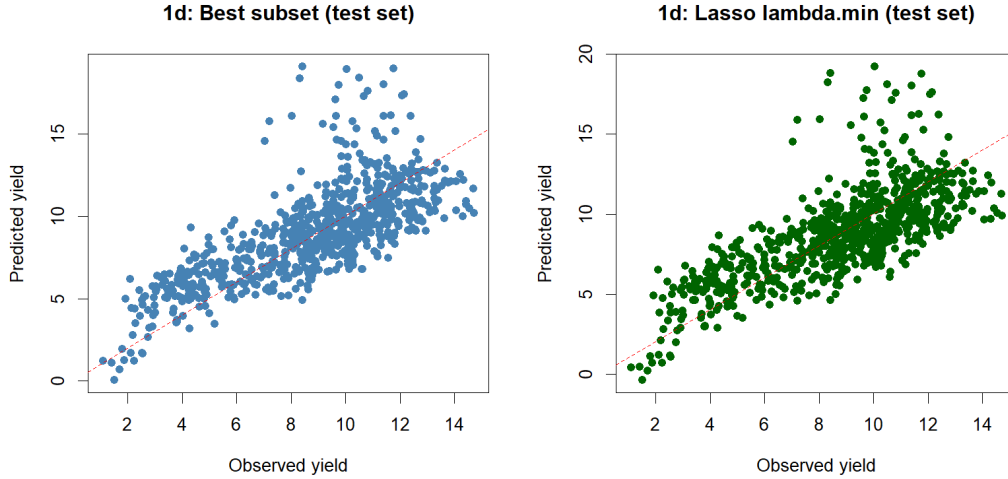


Figure 3: As Figure 2, for the year-parity split. Note the wider scatter around the 1:1 line.

The mechanism is the spatial dependence documented in Answer 1b. Under the random split, a test observation from, say, location 7 in a given year has training-set neighbours from other locations in the same year. Because temperature is correlated at $r = 0.96$ across locations within a year, those neighbours carry much of the information needed to predict the test point’s yield — the model interpolates rather than extrapolates. The year-parity split removes this channel: no year appears in both sets, so the model must predict under meteorological conditions it has never observed at any location. The resulting test R^2 of 0.45–0.49 is the honest estimate of out-of-sample performance.

3 European Climate Trends

3.1 Per-cell trend analysis: Answer 2a

The gridded climate data cover the European domain from approximately 35°N to 75°N and 10°W to 30°E, at roughly 0.25° resolution, over 76 annual timesteps from 1950 to 2025. Annual mean temperature ranges from -6.98°C to 24.73°C ; annual precipitation sums range from 0 to 4938.4 mm. Ocean cells are masked (NA).

For each land grid cell, a linear regression of the variable on year was fitted and the null hypothesis H_0 : slope = 0 was tested against the two-sided alternative at $\alpha = 0.05$. This gives 11,791 temperature tests and 11,918 precipitation tests.

The strongest warming trend is $0.058^\circ\text{C yr}^{-1}$, and the strongest cooling trend is $-0.012^\circ\text{C yr}^{-1}$. For precipitation, the strongest increase is 17.1 mm yr^{-1} and the strongest decrease is -7.8 mm yr^{-1} ; both are isolated single cells and likely artefacts of the underlying interpolation over complex terrain.

Of the 11,791 temperature cells, **11,512 (97.6%)** show a significant trend at $\alpha = 0.05$. For precipitation, **4,799 of 11,918 (40.3%)** are significant. Under a global null of no trends anywhere, we would expect approximately 589.6 and 595.9 false positives respectively. Both observed counts are far above this baseline, indicating genuine widespread signal.

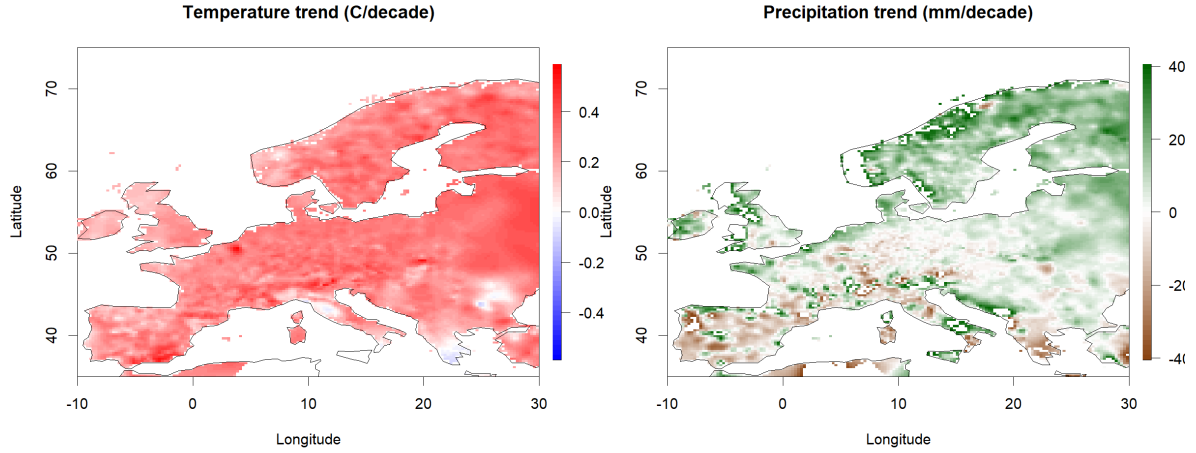


Figure 4: Linear trends in annual mean temperature (left, °C/decade) and annual precipitation sums (right, mm/decade) over Europe, 1950–2025. Colour scales are symmetric around zero.

Figure 4 shows the spatial distribution of the raw trends. Warming is coherent across nearly the entire domain, with the strongest signal over Scandinavia, the Baltic, and Central/Eastern Europe; cooling patches appear only in isolated parts of the British Isles and the southeastern Balkans. Precipitation trends are much patchier: increases dominate in Scotland, Norway, and the Alps, while decreases cluster in Iberia, Italy, and the western Balkans.

3.2 Multiple testing correction: Answer 2b

The Benjamini–Hochberg procedure was applied separately to each variable’s p -values, with a self-chosen false discovery rate of $q = 0.05$. This guarantees that, on average, no more than 5% of the rejected null hypotheses are false positives.

For temperature, the number of significant cells is essentially unchanged ($11,512 \rightarrow 11,510$; 0.02% reduction). For precipitation, the count falls from 4,799 to 3,838, a 20.0% reduction. The expected number of false discoveries among the surviving rejections is $q \times (\text{number of rejections})$: 575.5 for temperature and 191.9 for precipitation.

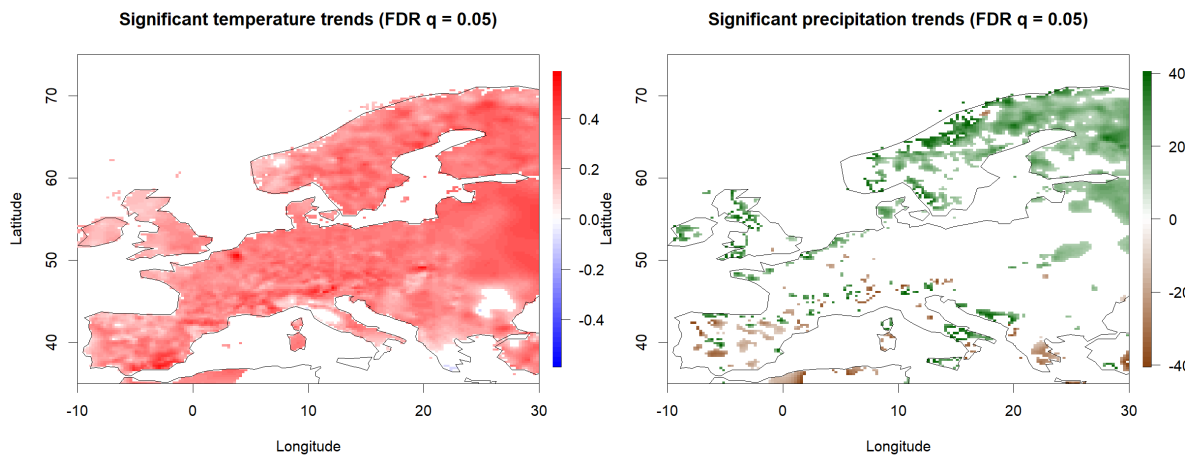


Figure 5: Grid cells with a statistically significant trend after Benjamini–Hochberg correction ($q = 0.05$).

After correction, the spatial patterns remain largely intact: temperature warming is still near-ubiquitous, and precipitation retains its regional structure (wetter across Scandinavia, drier across Iberia and the western Mediterranean). The divergence in correction behaviour reflects a real difference in signal-to-noise ratio. Temperature trends are large relative to year-to-year variability and spatially coherent, so almost every raw-significant cell corresponds to a genuine trend. Precipitation trends are smaller relative to interannual variability and spatially patchier, so a meaningful fraction of the raw-significant cells were likely noise — which the FDR correction removes.

4 Conclusion

Both analyses make the same methodological point from different directions: the assumptions underlying standard statistical learning procedures are frequently violated by Earth system data, and the consequences show up in the results. In the Iowa crop panel, ignoring spatial dependence via a random split yields a test R^2 of 0.63–0.64; respecting it via a year-parity split yields 0.45–0.49. The 0.13–0.19 gap in R^2 is the cost of the random split’s implicit information leakage, and it would not survive in operational forecasting, where future years are genuinely unseen. In the European climate grids, failing to correct for multiple testing inflates the count of significant trends by roughly 20% for precipitation, while barely affecting temperature, since the two variables differ in signal-to-noise ratio. In both cases, checking whether the data’s structure is compatible with the method’s assumptions — and choosing the variant that accounts for that structure explicitly — is what makes the reported performance and significance estimates trustworthy.